



COMPUTER NETWORKS

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Department of Computer Science and Engineering

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Transport Layer

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3.4 principles of reliable data transfer

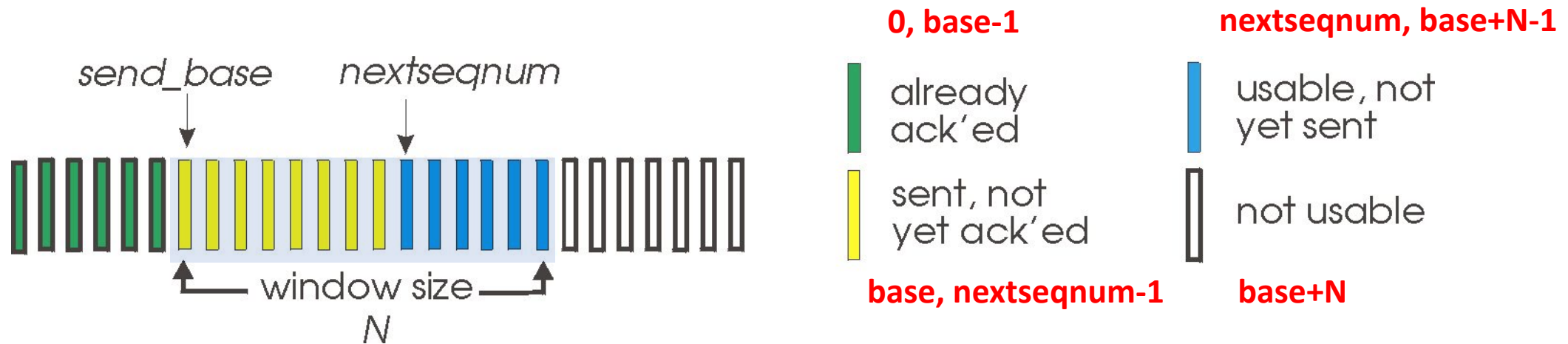
3.5 connection-oriented transport: TCP

- segment structure
- reliable data transfer
- flow control
- connection management

3.6 principles of congestion control

3.7 TCP congestion control

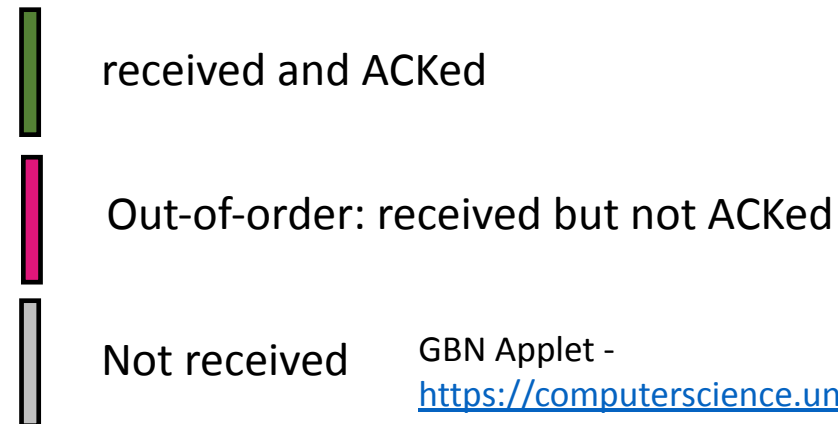
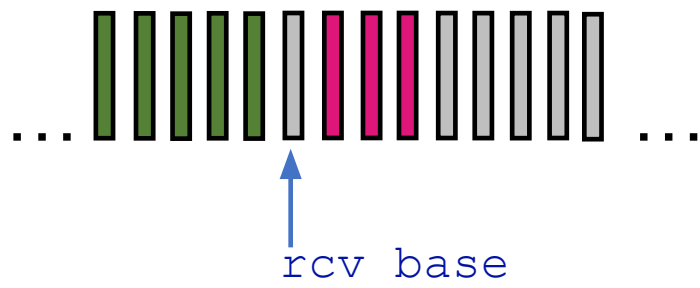
- sender: “window” of up to N , consecutive transmitted but unACKed pkts
 - k -bit seq # in pkt header



- **cumulative ACK:** $ACK(n)$: ACKs all packets up to, including seq # n
 - on receiving $ACK(n)$: move window forward to begin at $n+1$
- timer for oldest in-flight packet
- **timeout(n):** retransmit packet n and all higher seq # packets in window

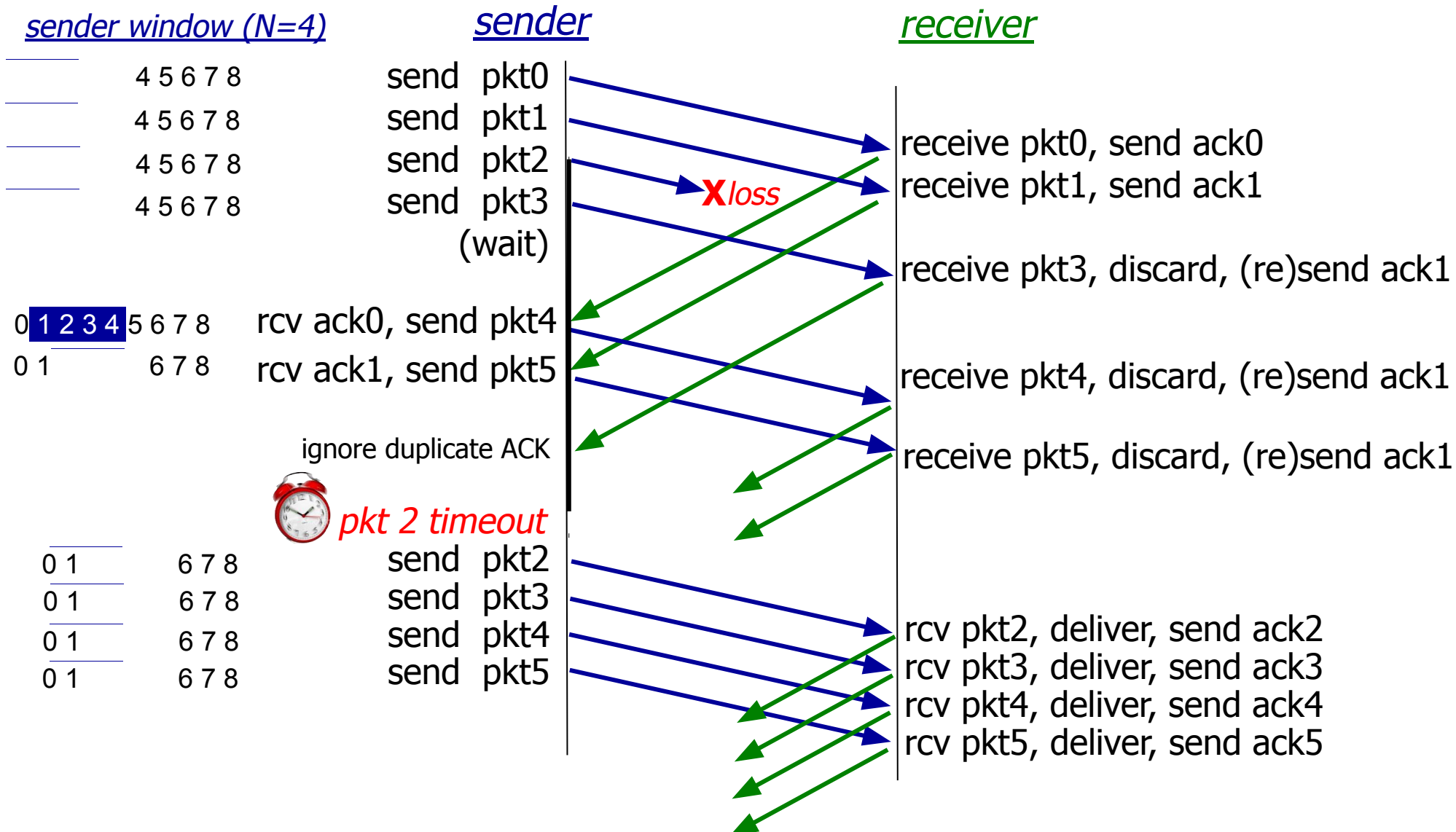
- ACK-only: always send ACK for correctly-received packet so far, with highest *in-order* seq #
 - may generate duplicate ACKs
 - need only remember `rcv_base`
- on receipt of out-of-order packet:
 - can discard (don't buffer) or buffer: an implementation decision
 - re-ACK pkt with highest in-order seq #

Receiver view of sequence number space:



GBN Applet -

<https://computerscience.unicam.it/marcantoni/reti/applet/GoBackProtocol/goback.html>



In GB3, if every 5th packet that is transmitted is lost and if we have to send 10 packets, then how many transmissions are required?

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THANK YOU

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